

### Research Summary

Doctoral researcher in Computer Science at Kennesaw State University working at the intersection of quantum communication, quantum cryptography, and quantum-augmented networks. Research centers on designing and analyzing protocols for secure quantum communication, including quantum key distribution, multi-photon and quantum secure direct communication protocols, and joint encryption with error correction, and on using quantum and post-quantum techniques to strengthen classical network and infrastructure security, from application-layer (HTTP) protocols to microgrid cybersecurity. This work pairs theoretical protocol design with simulation-based evaluation of noise, network topology, and practical attack models. Published in venues including Scientific Reports, Cryptologia, the Infocommunications Journal, and SPIE and IEEE proceedings; serves as a reviewer for several IEEE, ACM, and Springer-Nature journals.

### Research Interests

- 1 Secure Quantum Communication and Networking
- 2 Secure Quantum-Augmented Networks
- 3 Quantum ML/AI for Cybersecurity
- 4 Cyber-Physical Systems Security

### Teaching Interests

- 1 Cybersecurity and Network Security
- 2 Cryptography and Quantum Communication
- 3 Computer Networks and Data Communications

### Education

- 2023–Present **Ph.D. in Computer Science**, Kennesaw State University, Marietta, USA  
Current GPA: 3.85/4.00  
Tentative dissertation title: *Secured Quantum-Augmented Networks for Cyber-Physical Infrastructures*
- 2020–2023 **B.Sc. (Hons.)**, Ashoka University, Sonapat, India, *Major in Physics*  
Graduated with latin honors: *Cum Laude*. GPA 3.69/4.00

### Research Experience

- 2023–Current **Graduate Research Assistant**, Kennesaw State University, Marietta, USA, *Quantum networks and Communication*  
Conducting theoretical and simulation-based studies for different aspects of Quantum networks and communication.
- 2022–2023 **Research Assistant**, Ashoka University, Sonapat, India, *Simulating Micromagnetic Systems*  
Using OOMMF (C++) or MuMax3 (GoLang) to simulate the behavior of permalloy in varying magnetic fields.

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## Skills

### Technical Skills

Languages Python, MATLAB, LaTeX  
Frameworks IBM Qiskit, PennyLane

### Personal Skills

Languages English, Hindi

### Workshops

2026 BEACON Labs Cybersecurity Workshop

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## Services & Responsibilities

### Teaching

Formats Online and in-person instruction

- 2026 Instructor, online Quantum Cryptography course for graduate students: **Introduction to Cryptography and Its Applications**. Covered qubit fundamentals, QKD protocols, practical quantum networks, and open problems in quantum networking.
- 2026 Designed course slides and a Docker-based interactive lab on middlebox attacks for hands-on cybersecurity instruction.
- 2026 Produced a video series on TLS, blockchain security, and post-quantum cryptography.

### Student-Mentoring

I have mentored a total of **five students** in the past (3 M.Sc. and 2 undergraduate students), and currently mentoring **two students**: an undergraduate and a high-school working with our lab.

- 2026 Mentored a high-school student for a summer internship exchange program for research in multi-modal Steganography.
- 2026 Mentored two undergraduate students as part of the NSF-REU program for research in topics of Quantum-GANs for network security.
- 2023-24 Mentored two M.Sc. (CS) students with their respective theses on quantum communication protocols and developing an adaptive quantum education platform.
- 2024-26 Mentored an M.Sc. (CS) student with their thesis centered around evolving the quantum education platform by making it more modular and accessible to all students.

### Reviewer Roles

Reviewer **IEEE Transactions on Mobile Computing**  
Reviewer **Scientific Reports**  
Reviewer **IEEE Access**  
Reviewer **Advanced Quantum Technologies (QUTE)**  
Reviewer **ACM Transactions**  
Reviewer **Infocommunication Journal (ICJ)**  
Reviewer **Discover Computing, Springer Nature**  
Feature Editor **ACM XRDS**

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## Publications

### Journal Articles

- 2026 **Jha, N., Parakh, A., & Subramaniam, M.** (2026). An improved quantum anonymous notification protocol for quantum-augmented networks. *Frontiers in Quantum Science and Technology*, 5, Article 1704298. doi:10.3389/frqst.2026.1704298.

- 2026 **Jha, N.**, Parakh, A., & Subramaniam, M. (2026). Quantum key distribution: Bridging theoretical security proofs, practical attacks, and error correction for quantum-augmented networks. *Cryptologia*, 1–58. doi:10.1080/01611194.2025.2596885.
- 2025 **Jha, N.**, Parakh, A., & Subramaniam, M. (2025). Multi-photon QKD for practical quantum networks. *Infocommunications Journal*, 17(2), 72–82. doi:10.36244/ICJ.2025.2.9.
- 2024 **Jha, N.**, Parakh, A., & Subramaniam, M. (2024). Joint encryption and error correction for secure quantum communication. *Scientific Reports*, 14(1), Article 24542. doi:10.1038/s41598-024-75212-8.

### Conference Publications

- 2026 Paudel, P., **Jha, N.**, & Parakh, A. (2026). Evaluating multimodal steganalysis for split-payload audiovisual steganography. Accepted for presentation at *IEEE IMNS*. arXiv preprint arXiv:2606.08726. Preprint DOI: 10.48550/arXiv.2606.08726.
- 2026 Paudel, P., **Jha, N.**, Parakh, A., & Subramaniam, M. (2026). Hybrid quantum-classical GANs for the generation of adversarial network flows. In *Quantum Information Science, Sensing, and Computation XVIII*, Proceedings of SPIE, Vol. 14021, Article 140210B. doi:10.1117/12.3095130.
- 2026 **Jha, N.**, Paudel, P., Parakh, A., & Subramaniam, M. (2026). A novel quantum augmented framework to improve microgrid cybersecurity. In *Quantum Information Science, Sensing, and Computation XVIII*, Proceedings of SPIE, Vol. 14021, Article 1402109. doi:10.1117/12.3095119.
- 2025 **Jha, N.**, Parakh, A., & Subramaniam, M. (2025). Towards a quantum-classical augmented network. In *Quantum Computing, Communication, and Simulation V*, Proceedings of SPIE, Vol. 13391, pp. 72–86. doi:10.1117/12.3042150.
- 2024 **Jha, N.**, Parakh, A., & Subramaniam, M. (2024). An ML-based approach to quantum-augmented HTTP protocol. In *2024 IEEE International Conference on Quantum Computing and Engineering (QCE)*, Vol. 2, pp. 591–592. IEEE. (Poster.) doi:10.1109/QCE60285.2024.10420.
- 2024 **Jha, N.**, Parakh, A., & Subramaniam, M. (2024). Effect of noise and topologies on multi-photon quantum protocols. In *Quantum Computing, Communication, and Simulation IV*, Proceedings of SPIE, Vol. 12911, pp. 148–161. doi:10.1117/12.3000586.

### Preprints and Manuscripts Under Review

- 2026 **Jha, N.**, & Parakh, A. (2026). A novel steganography scheme using quantum Hilbert transform. arXiv preprint arXiv:2609.07894. Preprint DOI: 10.48550/arXiv.2609.07894. Manuscript under review at *Multidimensional Systems and Signal Processing*, special issue on Signal Processing for Quantum Technologies.
- 2026 **Jha, N.**, Paudel, P., & Parakh, A. (2026). Quantum-assisted session authentication in quantum-augmented networks. Manuscript under review at *Academia Quantum*.

### General Audience Articles

- 2026 **Jha, N.**, & Parakh, A. (2026). Toward a global quantum Internet: A review of challenges facing aerial quantum networks. *IEEE Potentials*. doi:10.1109/MPOT.2026.3657057.
- 2026 **Jha, N.**, & Parakh, A. (2026). Are quantum voting protocols practical? arXiv preprint arXiv:2601.17514. Preprint DOI: 10.48550/arXiv.2601.17514.
- 2026 **Jha, N.**, & Parakh, A. (2026). Quantum error correction: A solution to all of our quantum worries! Manuscript under review at *IEEE Potentials*.